

# Rohan Mehra

Graduate Student in Computer Science

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## Education

- 2025 – present **ISIMA, Université Clermont Auvergne** (Ranked 8/34)  
*MS in Computer Science*
- 2021 – 2025 **Indian Institute of Science Education and Research (IISER) Bhopal** (Grade: 8.92/10)  
*BS in Electrical Engineering and Computer Science*

## Research Experience

- Mar. – present **LIFAT, CNRS, Université de Tours, M2 Intern**
- 2026
- Advised by Prof. Gilles Venturini as part of RFAI team.
  - 📄 ○ Working on AI-assisted image recognition of bumblebees (*bombus*).
  - Using transfer learning, and explainable AI approaches.
- 2025 – 2026 **Institut Pascal, CNRS, Université Clermont Auvergne, M2 Project Student / Intern**
- & May – July
- 2024
- Advised by Dr. Mathieu Labussière as part of ComSee team.
  - 📄 ○ Worked on deep learning with SWIR sensor for road perception in adverse weather.
  - 📄 ○ Models worked on include Grounded SAM, YOLO, RF-DETR [Sem. Segmentation] and CUT, CycleGAN, Pix2Pix, BBDM [GANs & diffusion-based]
- Jan. – April **IISER Bhopal, BS Project Student**
- 2025
- Advised by Dr. Arpit Sharma at the Dept. of EECS.
  - 🔗 📄 ○ Worked on using Zero-Shot Learning model for Sector Classification in CrowdRE data.
  - Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow
- May – July **Indian Institute of Space Science and Technology, Intern**
- 2023
- 📄 ○ Advised by Prof. Deepak Mishra as part of the CVVR Lab
  - 📄 ○ Worked on robotics and implementation of SLAM through Kalman filter Methods.
  - Used Robot Operating System (ROS), RViz, and Gazebo.

## Selected Publications

Full list available on [Google Scholar](#)

- 🔗 **Rohan Mehra**, Alexandre Riffard, Yannis Loumouamou, Mathieu Labussière, “Analysis of SWIR Imaging Detection Performance Under Adverse Environmental Conditions for Autonomous Driving Systems”, **Under review at IEEE Transactions on Intelligent Vehicles (T-IV)**, 2026.
- 🔗 Yannis Loumouamou, Alexandre Riffard, **Rohan Mehra**, Mathieu Labussière, “Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection”, *Reconnaissance des Formes, Image, Apprentissage et Perception (RFIAP)*, 2026.
- 🔗 **Rohan Mehra**, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, “Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.”, *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025. [Oral]

## Teaching Experience

- 2024 & 2023 **Basic Electronics, IISER Bhopal (× 2)**
- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
  - Conducted tutorial sessions (1 hour/week) and prepared assignment solutions.

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## Academic Service

Reviewer IEEE Sensors Journal (2026)

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## Minor Course Projects

- 2025 **Risk-Aware Multi-Epoch Package Delivery Planning** *Optimization @ IISER-B*  
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  - Used greedy algorithm to solve package delivery problems and built a GUI-based visualization tool.
  - Developed a GUI-based visualization tool using Tkinter and Matplotlib.
- 2024 **Training Multi-Robot Warehouse (RWARE) Environment** *RL @ IISER-B*  
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  - Trained robots for optimized coordination in the RWARE environment.
  - Employed Deep Q-Network (DQN) for efficient learning.
- 2024 **Implementation of Grover's Algorithm** *Quantum Computing @ IISER-B*  
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  - Implemented Grover's Algorithm for the 3-SAT problem demonstrating quantum speedup.
  - Utilized PennyLane for designing quantum circuits.

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## Technical Skills

Languages Python (with PyTorch framework), C++, C, Java  
Software ROS, Gazebo, MATLAB, Git,  $\LaTeX$ , Wolfram Mathematica

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## Events

- Mar. 2026 **ELLIS Winter School on Foundation Models (FoMo)**  
Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants (~250 participants).
- Feb. 2026 **Researchers' Day – DATA Program I-SITE CAP 20-25**  
Presented a poster at PhD day, Clermont-Ferrand (sole master student).

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## Fellowships / Achievements

- 2025 **MIAI Cluster Excellence Scholarship**  
Selected to receive a scholarship offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to ~20 candidates university wide.
- 2024 **IASc-INSANA-SASI Summer Research Fellowship** *[Declined]*  
Selected for research internship fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top ~5% of applicants.
- 2021 **Indian Maritime University Undergraduate Admissions**  
Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

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## Positions of Responsibility / Extra Curriculars

- 2023 – 2024 **Cultural Secretary, IISER Bhopal**  
Elected representative of the student body for organizing all cultural events at IISER Bhopal. Led a team of 70 performers to the Inter IISER Cultural Meet 2023 held at IISER Mohali.
- 2022 **NCC 'B' Certificate, National Cadet Corps, India**  
Completed 'B' division NCC military training from 8 UP BN NCC, Aligarh.

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## Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)  
Hobbies Table Tennis, Quizzing